

10-Mark Questions & Answers — MPCA Module 6: Pentium 4 & ARM Processor

Q1. Explain the evolutionary comparison of Intel processors (8086 to Pentium 4), Pentium 4 NetBurst architecture, Hyper-Threading, and ARM RISC processor architecture in detail.

Comprehensive 10-mark master answer covering:

1. Comparative analysis of 8086, 80386, Pentium, Pentium II, and Pentium 4.
2. Pentium 4 NetBurst components: 20-stage pipeline, Trace Cache, Rapid ALUs, Quad-Pumped FSB.
3. Hyper-Threading simultaneous multithreading architecture.
4. ARM Processor architecture, Load-Store paradigm, Barrel Shifter, 37 registers, ARM/Thumb modes, and mobile dominance.